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Personalised Driver Risk Assessment With Adaptive Feedback Using Crowdsensed Telemetric Data

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ABSTRACT

This paper presents a comprehensive, data-driven framework for personalised driving risk assessment, designed to enhance driver safety within intelligent transportation systems. By leveraging crowdsensed telemetric and road environment data, the framework captures diverse driving behaviours and contextual factors to provide real-time, individualised risk insights. The two-phase framework combines Gaussian Mixture Model (GMM) clustering, Deep Embedded Clustering (DEC), and Fully Connected Network (FCN) for accurate risk classification and prediction, while Deep Q-Learning (DQN) delivers adaptive feedback that encourages safer driving practices. Extensive evaluation shows that our approach outperforms traditional models in both accuracy and adaptability with an accuracy score of 95% and an average F1-score of 0.94, demonstrating its value in capturing complex driver behaviour patterns and contributing a scalable solution for transportation safety.

1 | Introduction

The increasing complexity of urban traffic systems and the rapid development of Intelligent Transportation Systems (ITSs) and autonomous vehicles have brought unprecedented opportunities for enhancing road safety and efficiency. Traditional traffic risk models, often generalised and static, fail to account for individual driving behaviours and the nuanced factors that influence traffic safety, such as environmental and road conditions [1]. With the advent of advanced telematics and machine learning, there is growing potential to transform risk prediction from generalised models to highly personalised systems that adapt to individual drivers' unique behaviour patterns. In particular, personalised traffic risk assessments have emerged as a promising approach to providing driver-specific feedback, potentially improving both driver awareness and safety on the road [2]. This work contributes to the expanding field of Driver Feedback Systems for Efficiency and Safety by offering a framework that delivers real-time,

personalised guidance, enhancing drivers' ability to recognise and mitigate risky behaviours.

However, implementing a personalised risk assessment system is fraught with challenges. Accurate prediction requires the integration of diverse data sources, including telemetric data, road network information, and environmental conditions, each with distinct spatial and temporal characteristics [3]. Additionally, driver behaviours are inherently variable and may overlap across different risk profiles, complicating the segmentation process. The shortage of sufficiently labelled data further compounds these challenges, as supervised models rely heavily on annotated examples to learn accurate patterns [4, 5]. Without ample labelled data, capturing the full spectrum of driver behaviours and ensuring precise risk classification becomes difficult. This growing data complexity necessitates advanced clustering and classification methods to extract meaningful patterns while ensuring model scalability and interpretability. Addressing these

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challenges is essential, particularly as ITS and autonomous vehicles become more prevalent, demanding real-time, adaptive frameworks capable of managing dynamic road environments.

To address the challenges associated with the increasing demand for personalised driver risk assessment within ITSs, this paper introduces a comprehensive, cost-effective, data-driven framework that leverages crowdsensed vehicle telemetric and road environment data to deliver individualised risk assessments for drivers. Specifically, our contributions are:

We propose a data-driven framework for personalised driving risk assessment, incorporating spatiotemporal data fusion and multidimensional feature engineering to derive insights from driving behaviour, road conditions, and environmental factors.

We present a two-phase framework that integrates Gaussian Mixture Model (GMM) clustering with Deep Embedded Clustering (DEC) for accurate classification of risk profiles while employing a Fully Connected Network (FCN) for predictive risk modelling. Additionally, we incorporate Deep Q-Learning (DQN) to provide adaptive, real-time feedback that enables drivers to make immediate behavioural adjustments.

We evaluate our framework across three major United States cities, Philadelphia (PA), Atlanta (GA), and Memphis (TN), using a crowdsensed telemetric dataset combined with road environment data. The results demonstrate that our framework accurately predicts driving risk levels, achieving accuracy, precision, and recall rates all exceeding 90%, thereby outperforming baseline methods.

By synthesising advanced machine learning techniques with multi-source data, this framework offers actionable, personalised insights that promote driver safety and adaptability, representing a scalable solution for applications in transportation and ITSs.

The remainder of this paper is organised as follows. Section 2 provides a review of related work in personalised driver risk assessment. Section 3 introduces the overall framework and preliminaries. Section 4 explains the driving behaviour profiling phase. Section 5 describes the personalised driver risk assessment phase. Section 6 presents the evaluation of the framework, and Section 7 concludes the paper with a summary of contributions and insights.

2 | Related Works

2.1 | Personalised Traffic Risk Prediction

Recent advancements in personalised traffic risk prediction have shifted focus towards integrating individual driving behaviours and environmental factors to improve road safety. Traditional models, which treat all drivers similarly, are being replaced by approaches that tailor risk assessments to specific driver behaviours using telematics data, leading to more precise and relevant safety measures. For instance, a study introduced a predictive framework that considers specific behaviours, such as acceleration patterns and braking tendencies, to forecast accident risks [6]. Similarly, it has been demonstrated that deep

learning techniques applied to telematics data can outperform conventional models in predicting traffic incidents [7], highlighting the effectiveness of personalised approaches. A significant advancement in this field is personalised lane-changing risk prediction. Using driving volatility indices and dynamic clustering, researchers have categorised drivers in real time, finding that aggressive drivers are at higher risk during lane changes [8]. The Light Gradient Boosting Machine (LightGBM) algorithm has proven effective in predicting these risks, offering the potential for Advanced Driver-Assistance Systems (ADAS) to provide more targeted safety interventions [9].

Comprehensive reviews of traffic accident prediction studies emphasise the importance of machine learning and diverse data sources [10]. The integration of real-time data is critical for improving the accuracy and timeliness of risk predictions. Meta-feature-based models have also emerged, enhancing the prediction of accident severity and incidence by leveraging various data features [11]. By leveraging a variety of data features, these models aim to enhance prediction accuracy, offering a promising avenue for more effective traffic safety interventions. Furthermore, personalised trajectory-based risk prediction has shown promise in reducing lane departure accidents, particularly on curved roads. These models contribute to more tailored and effective road safety measures by considering individual driving patterns [12]. However, a gap remains in integrating personalised models with real-time contextual information, such as weather and road conditions, which are crucial for accurate risk predictions. Addressing this, our framework aims to incorporate both driver-specific behaviours and contextual factors, resulting in a more holistic approach to traffic risk prediction.

2.2 | Driver Behaviour Modelling

Driver behaviour modelling is essential for Advancing Autonomous Vehicles (AVs), enhancing traffic management, and improving driver-assistance technologies. By predicting and understanding driver actions, these models contribute to safer and more efficient transportation systems. Various methods have been developed to model and simulate driver behaviour, such as Inverse Reinforcement Learning (IRL), which has gained traction as a robust method that learns driving policies from observed behaviours [13]. Another notable variant is Adversarial Inverse Reinforcement Learning (AIRL), which reconstructs reward structures from driver actions, allowing generalisation to new scenarios [13]. However, IRL is computationally demanding, posing challenges for real-time applications. To manage diverse environmental conditions, domain adaptation techniques create invariant representations across factors such as lighting or sensor changes [14]. While these models enhance reliability in varied settings, they sometimes struggle with unanticipated environmental shifts.

Traditional statistical and regression models remain relevant, especially for interpretable analysis in specific conditions. For instance, Tobit and mixed-effects regression models have been applied to braking and steering behaviour on highways, revealing factors like speed and road curvature that influence driver response [15]. Yet, these models may not capture the non-linear dynamics required for more complex driving scenarios. Advanced

game-theoretic models also play a role by simulating multi-agent interactions, such as decision-making at intersections, though they often rely on assumptions of rationality that may not reflect real-world behaviour [16]. Neural Networks, particularly Recurrent Neural Networks (RNNs), have proven effective for modelling sequential driving data. These models can capture changes in driving patterns over time, making them useful in scenarios like car-following. However, their lack of interpretability can hinder applications requiring insights into decision-making processes [17]. Additionally, Human Performance Models (HPMs) further focus on quantifying human factors in driving, such as interactions with automation, but their reliance on predefined human metrics can restrict generalisability across diverse driver populations [18].

These models have wide-ranging applications, including Advanced Driver Assistance Systems (ADASs), which provide timely warnings, and Traffic Simulation, which supports urban planning by predicting congestion patterns [19]. In autonomous vehicles, driver behaviour models are crucial for predicting human actions, facilitating safer navigation in mixed environments. Personalised driver support systems also use these models to deliver individualised alerts, enhancing user experience by reducing unnecessary warnings. While these models have significant contributions, limitations still exist related to scalability, computational efficiency, and interpretability. Methods like IRL and game-theoretic models can be computationally intensive, limiting real-time use [20]. Neural networks, while powerful in capturing complex behaviour, often lack transparency, and traditional statistical models can oversimplify real-world driving dynamics [21]. Most importantly, existing studies tend to focus on general trends without delivering personalised, adaptive feedback that evolves with individual driving patterns.

2.3 | Feature Engineering for Driving Behaviour Assessment

Feature engineering is crucial for driving behaviour assessment, transforming raw telematics data such as signals, GPS, accelerometers, OBD-II, and vehicle CAN bus data into interpretable metrics that support applications ranging from usage-based insurance to real-time safety alerts. At its core, this process involves extracting event-based features like counts or flags for harsh acceleration or deceleration exceeding 3m/s^2 , a threshold supported by several studies [22, 23], pattern-based summaries such as rolling statistics of speed variance and acceleration to capture consistent driving styles, and contextual variables including road type, weather, and traffic conditions [24, 25]. Empirical studies highlight the effectiveness of these methods. One study demonstrated that acceleration-derived features can classify aggressive driving with nearly perfect accuracy [26], while another study utilised historical feature sets to predict accident risk with high precision [27]. Moreover, a study achieved 93.3% accuracy in driver profiling using pattern-based motion sensor features [28], while another study showed that integrating contextual factors significantly enhances the prediction of risky events [29]. By combining these diverse features, models can thoroughly assess driving behaviour, advancing safety, efficiency, and tailored services.

3 | Preliminary and Framework Overview

In this section, we define several terms used in this paper and present an overview of the framework.

3.1 | Preliminary

Definition 1 Telemetric Data: Telemetric data encompasses vehicle-generated information, including speed (v), acceleration (a), geographic coordinates (latitude λ , longitude ϕ), and timestamps (t). Each data record $T = \{t_i, v_i, a_i, \lambda_i, \phi_i\}$ represents a snapshot of the vehicle's state, capturing both temporal and spatial aspects of driving behaviour.

Definition 2 Spatiotemporal Data Fusion: Spatiotemporal data fusion is the process of integrating datasets that have both spatial (location-based) and temporal (time-based) attributes, combining telemetric data T_i with road network information R_j and weather conditions W_k . This fusion allows for a comprehensive analysis of how both immediate and environmental factors influence driving behaviours.

Definition 3 Driving Behaviour Profiling: Driving behaviour profiling categorises drivers based on features derived from telemetric data to assess risk tendencies.

Definition 4 Risk Classification: Risk classification involves assigning each driver a risk level based on their behaviour profile. Using a FCN, risk categories are predicted based on extracted feature vectors $X = \{x_1, x_2, \dots, x_n\}$.

3.2 | Framework Overview

The proposed framework for personalised driving risk assessment integrates data fusion, behavioural analysis, and machine learning-driven classification to provide drivers with actionable safety insights. The framework is organised into two main phases: driving profiling and personalised driving risk Assessment, as shown in Figure 1. These phases are designed to capture detailed driver behaviour profiles and deliver tailored risk feedback.

3.2.1 | Phase 1: Driving Profiling

In the driving profiling phase, multiple data sources and clustering techniques are combined to build individualised driver profiles. Spatiotemporal data fusion merges crowdsensed telemetric data with road environment and temporal data, creating a context-rich dataset that reflects driving behaviours under varied conditions. In the second step, the framework applies multi-dimensional feature engineering to extract insights from the fused data, capturing diverse aspects of driving patterns and behaviours. This feature engineering step leverages aggregated data to model both typical and unique driving traits across different conditions and regions. Finally, driving profiling with GMM clustering segments drivers into behaviour-based clusters using GMM clustering, which assigns drivers probabilistically to clusters, acknowledging the overlapping nature of real-world driving styles and establishing a foundation for personalised risk assessment.

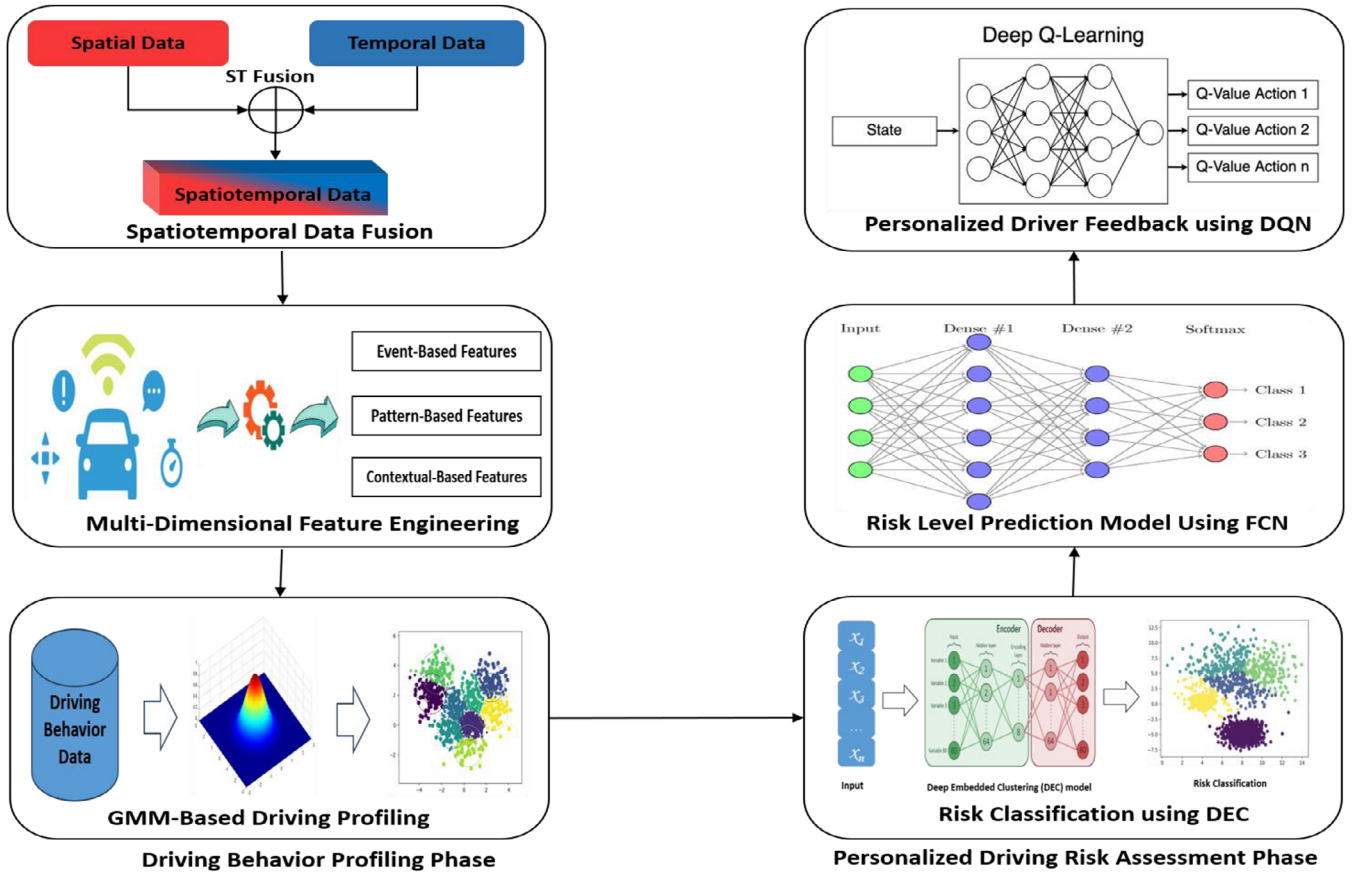


FIGURE 1 | Personalized driver risk assessment framework overview.

3.2.2 | Phase 2: Personalised Driving Risk Assessment

Building on the profiles from Phase 1, the Personalised Driving Risk Assessment phase classifies and evaluates driver risk levels to provide individualised insights. Risk classification with DEC utilises DEC to capture complex, non-linear patterns within the driver data, enhancing the accuracy of risk categorisation and distinguishing nuanced risk profiles. The Risk Level Prediction Model employs an FCN to assign risk scores to drivers based on their profiles, ensuring real-time, reliable predictions. Lastly, Personalised Driver Feedback with Reinforcement Learning uses DQN to deliver adaptive, real-time feedback, continuously adjusting based on driver responses to encourage safer driving practices.

This two-phase framework combines data fusion, multidimensional feature engineering, advanced clustering, predictive modelling, and adaptive feedback, providing a robust, scalable solution for enhancing driver safety within ITSs.

4 | Driving Behaviour Profiling

4.1 | Spatiotemporal Data Fusion

We integrated a large-scale crowdsourced vehicle telemetric dataset with road environment and weather data through spatiotemporal fusion. Initially, we preprocessed the telemetric, weather, and road segment datasets to ensure operational effi-

ciency and data integrity. Subsequently, these datasets were fused using a spatiotemporal feature-level approach, combining spatial and temporal dimensions based on common attributes as shown in Figure 2. This process synthesises their respective feature vectors into a cohesive and enriched dataset.

Spatial fusion is conducted through geometric operations that map telemetric data to corresponding road segments, thereby incorporating essential spatial context into the analysis. Specifically, we define spatial join operations by transforming each telemetry record's longitude and latitude into a geometric point $P(\lambda_i, \phi_i)$, where λ_i and ϕ_i denote the longitude and latitude of the i -th telemetry record, respectively. We then first map each telemetry record to its nearest road segment as;

$$T_{\text{geom}} = \{P(\lambda_i, \phi_i) | t_i \in T\}, \quad (1)$$

where t_i is the timestamp of the i -th telemetry record and T denote the telemetry record, while the road segment data is denoted as

$$R_{\text{geom}} = \{P(\lambda_j^E, \phi_j^E) | r_j \in R\}, \quad (2)$$

where r_j is the index of the j -th road segment and (λ_j^E, ϕ_j^E) its endpoint coordinates. The spatial join J of telemetric data with road segment data is formally expressed as:

$$J_S = \{t_i, r_j\} | d(P(\lambda_i, \phi_i), P(\lambda_j^E, \phi_j^E)) < \varepsilon \quad (3)$$

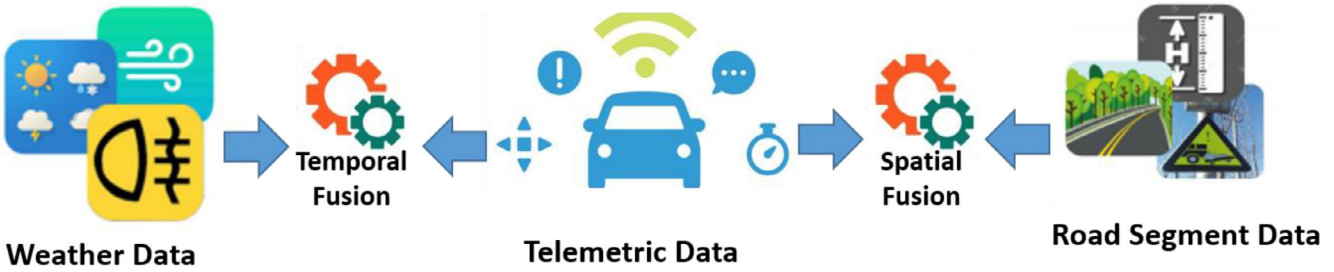


FIGURE 2 | Spatiotemporal data fusion.

where $d(\cdot, \cdot)$ is the chosen distance metric (geodesic distance) and ϵ the maximum allowable proximity threshold. This operation effectively associates each telemetric data point with its corresponding road segment, facilitating the integration of road attributes with driving events.

Temporal fusion aligns records based on their timestamps by defining a temporal join J_T between telemetric and weather data. We align each spatially matched pair $(t_i, r_j) \in J_S$ with its nearest weather observation. Denote the set of weather records as:

$$W = \{(w_k, t_k^w) \mid k = 1, \dots, K\}, \quad (4)$$

where w_k is the weather measurement taken at time t_k^w . The temporal join is defined as:

$$J_T = \{(t_i, r_j, w_k) \mid (t_i, r_j) \in J_S, k = \text{argmin} \mid \text{timestamp}_i - t_k^w \mid, \ell\} \quad (5)$$

where timestamp_i is the timestamp from t_i . Each tuple (t_i, r_j, w_k) pairs the telemetry record t_i with road segment r_j and the weather reading w_k whose timestamp is closest to timestamp_i . This formulation ensures that each telemetry record is paired with the temporally closest weather data, enabling a comprehensive analysis of how environmental conditions influence driving behaviour at specific moments.

Finally, for each $(t_i, r_j, w_k) \in J_T$, we construct a fused feature vector by concatenating the telemetry measurements m_i from t_i , the attributes of road segment r_j , and the weather variables w_k :

$$F = \{f_i \mid f_i = \text{concat}(m_i, \text{attrs}(r_j), w_k), (t_i, r_j, w_k) \in J_T\}. \quad (6)$$

Where m_i is the vector of telemetric measurements at time t_i , $\text{attrs}(r_j)$ denotes the set of road-segment attributes for segment r_j , and w_k is the vector of weather variables for record k . This concatenation yields f_i , an enriched feature vector combining vehicle behaviour, spatial context, and environmental conditions for each telemetry record.

4.2 | Multi-Dimensional Feature Engineering for Behaviour Extraction

We extracted driving behaviour features from the fused crowd-sensed telemetric data by adopting the method proposed by [30]. Specifically, we extracted three types of driving behaviours

using a multidimensional feature engineering approach: event-based features capture specific driving events indicative of risky behaviour; pattern-based features analyse recurring patterns over time, focusing on variability and consistency in driving; and contextual features integrate environmental and situational factors that might influence driving behaviour. The goal is to enhance the model's predictive power by capturing various aspects of driving behaviour and the surrounding environment. We selected event-based features because they focus on identifying specific driving events indicative of risky behaviour, such as sudden braking, harsh acceleration, or phone distractions [25]. These features are critical for real-time monitoring and intervention, enabling the detection of immediate safety risks during a driving session. By including event-based features, our approach identifies critical incidents that may require immediate corrective action, aligning with the goals of ADASs [31]. Additionally, pattern-based features examine recurring patterns in driving behaviour over time, focusing on the consistency and variability of driving habits, such as acceleration, braking frequency, and steering patterns [25, 32]. These features enable long-term driver behaviour profiling, which is valuable for applications like usage-based insurance and driver training. By incorporating pattern-based features, our methodology captures the temporal dynamics of driving, providing insights into habitual behaviours that influence safety and efficiency. Moreover, contextual features incorporate environmental and situational factors that may influence driving behaviour, such as road type, weather conditions, traffic density, and location-specific attributes [33]. These features ensure that driving behaviour is evaluated within the context of external conditions, enhancing the model's ability to adapt to situational variability. By including contextual features, our approach ensures a holistic analysis that adapts to external influences, improving the applicability of our model in real-world scenarios [25]. After engineering these features, we merged them into the core spatiotemporally fused dataset described in Section 4.1. This integration resulted in a single, comprehensive dataset that captures both the dynamic behaviour of the driver and the static and contextual factors influencing that behaviour.

Event-Based Features: Event-based features are designed to capture specific driving behaviours known to contribute to traffic risk. These include discrete, high-intensity actions such as harsh braking (where deceleration exceeds -3m/s^2) and rapid acceleration (where acceleration exceeds 3m/s^2), which have been widely adopted in the literature as thresholds indicating aggressive or unsafe manoeuvres and

overspeeding (where speed exceeds the legal limit for a given road segment). For instance, a study established $\pm 3\text{m/s}^2$ as a suitable boundary for detecting sudden longitudinal movements in smartphone-based telematics data, noting that safe acceleration events rarely exceed this level in naturalistic driving conditions [22]. Similarly, another study selected a 3m/s^2 cutoff after empirical inspection of real-world acceleration signals and reference to driver comfort literature [23]. Furthermore, one more study adopted a 3.0m/s^2 threshold in simulator-based safety studies to flag abrupt braking behaviour that departs from typical driving norms [34].

Harsh Braking: This feature identifies instances where the vehicle experiences sudden and significant deceleration. Harsh braking is detected when the deceleration exceeds a predefined threshold $a_{\text{threshold}}$. The equation is:

$$\text{harsh_braking}(t) = \int_0^1 \frac{if \frac{\Delta v(t)}{\Delta t}}{\text{otherwise}} < -a_{\text{threshold}} \quad (7)$$

where $\Delta v(t) = v(t) - v(t - \Delta t)$ is the change in velocity over the time interval Δt , and $a_{\text{threshold}}$ is the predefined threshold for deceleration. Here, Δt represents the sampling interval used for calculating the rate of change in velocity. The temporal dependence is introduced by considering the velocity at the current time step $v(t)$ and the velocity at the previous time step $v(t - \Delta t)$.

Rapid Acceleration: Rapid acceleration is identified when the vehicle's acceleration exceeds a specific threshold $a^+_{\text{threshold}}$. The equation is:

$$\text{rapid_acceleration}(t) = \int_0^1 \frac{if \frac{\Delta v}{\Delta t}}{\text{otherwise}} > a^+_{\text{threshold}} \quad (8)$$

where $\frac{\Delta v}{\Delta t}$ is the rate of change of velocity over time, and $a^+_{\text{threshold}}$ is the predefined threshold for acceleration.

Overspeeding: This feature identifies instances where the vehicle exceeds the legal speed limit for the road segment. It is calculated as:

$$\text{overspeeding}(t) = \int_0^1 \frac{if v(t)}{\text{otherwise}} > v_{\text{limit}} \quad (9)$$

where $v(t)$ is the vehicle's speed at time t , and v_{limit} is the speed limit of the road segment.

Pattern-Based Features: Pattern-based features are derived from time series analysis of the telemetric data to identify

recurring patterns. These features include speed variance, the moving average of speed over a defined window, and the rolling standard deviation of speed. These metrics are calculated to capture the variability and consistency of driving behaviour over time.

Speed Variance per Trip: This feature measures the variability in the vehicle's speed during a trip, indicating how much the speed fluctuates. It is calculated as:

$$\sigma^2_{\text{speed}} = \frac{1}{N} \sum_{i=1}^N (v_i - \bar{v})^2 \quad (10)$$

where v_i is the speed at time i , $\bar{v}$ is the mean speed over the trip, and N is the number of observations in the trip.

Rolling Acceleration Mean: This feature calculates the mean acceleration over a rolling window, providing insights into how the vehicle's acceleration changes over time. It is given by:

$$\mu_{\text{acceleration}}(t) = \frac{1}{W} \sum_{i=t-W+1}^t a_i \quad (11)$$

where W is the window size, t is the current time step, and a_i is the acceleration at time i .

Rolling Acceleration Standard Deviation: This feature measures the standard deviation of acceleration within a rolling window, indicating the variability in acceleration over time. It is calculated as:

$$\sigma_{\text{acceleration}}(t) = \sqrt{\frac{1}{W} \sum_{i=t-W+1}^t (a_i - \mu_{\text{acceleration}}(t))^2} \quad (12)$$

where W is the window size, t is the current time step, a_i is the acceleration at time i , and $\mu_{\text{acceleration}}(t)$ is the rolling mean acceleration.

Turning Rate: This feature measures the rate of change in bearing between consecutive data points, indicating sharp turns. Bearings are derived from GPS coordinates using the geodesic bearing formula:

$$\text{turning}_{\text{rate}} = |\text{bearing}_t - \text{bearing}_{t-1}| \quad (13)$$

where bearing_t is the bearing at time (t) , bearing_{t-1} is the bearing at the previous time step, and $|\cdot|$ is the absolute value, ensuring the turning rate is non-negative, reflecting the magnitude of the angular change. The telemetric datasets used have pre-calculated bearings for each coordinate.

Stop-and-Go Behaviour: This feature is detected when the vehicle's speed falls below 5 km/h and then increases:

$$\text{stop_and_go} = \int_0^1 \frac{\text{if speed} > \frac{5\text{km}}{h} \text{ and speed}_{\text{change}} > 0}{\text{otherwise}} \quad (14)$$

Contextual Features: Contextual features provide additional information based on the environment in which the vehicle is operating. These features include the extraction of temporal context, such as the time of day from timestamps, and the one-hot encoding of categorical variables like road types, enabling the model to differentiate between various driving environments. Additionally, weather conditions are encoded, allowing the model to account for specific environmental factors during prediction.

Night Driving: Trips occurring before 6 AM or after 6 PM are flagged as night driving:

$$\text{is_night} = \int_0^1 \frac{\text{if hour} < 6 \text{ or hour} > 18}{\text{otherwise}} \quad (15)$$

Rainy Driving: Driving in rainy conditions is detected based on precipitation data (PRCP):

$$\text{is_rainy} = \int_0^1 \frac{\text{if PRCP} > 0}{\text{otherwise}} \quad (16)$$

Weekend Driving: Driving on Saturday or Sunday is flagged as weekend driving.

4.3 | GMM-Based Driving Profiling

GMMs are probabilistic clustering algorithms that assume data points are generated from a mixture of several Gaussian distributions with unknown parameters. In the context of driver behaviour analysis, GMMs are particularly advantageous because they can capture complex, overlapping patterns in driving data, where behaviours may not fall into discrete categories but instead exist along a spectrum [35]. This flexibility aligns with the real-world variability observed in driving styles, such as transitions between aggressive and cautious behaviours under different road or environmental conditions.

Traditional clustering methods like k-means enforce hard assignments, where each data point belongs to exactly one cluster [36]. However, driving behaviours often exhibit ambiguity; for instance, a driver might demonstrate moderate acceleration patterns in urban areas but aggressive braking on highways [37]. GMMs address this issue by assigning probabilistic memberships, allowing a single trajectory to partially belong to multiple clusters. This capability is essential for capturing nuanced interactions between driver actions and contextual factors. Similarly, DBSCAN struggles with density overlaps that are common in driving data, failing to manage the variable density of driving data, where high-risk events are sparse compared to routine driving [38].

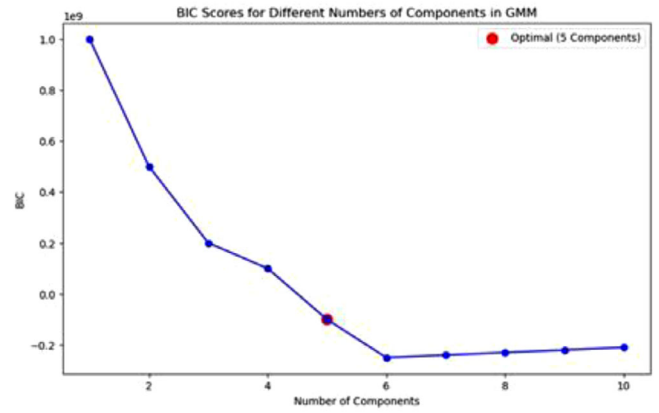


FIGURE 3 | BIC scores of different components.

Through GMM, each trajectory is assigned a probability of belonging to each cluster, and drivers are categorised into distinct behavioural profiles based on the highest probability cluster assignment. This probabilistic approach offers flexibility in defining clusters, reflecting the complexity and variability of driving behaviours in the real world. To determine the optimal number of components (clusters) for the GMM, we utilised the Bayesian Information Criterion (BIC), a widely used method for model selection in GMMs. The term 'components' refers to the number of Gaussian distributions (clusters) used to model the data, with each component representing a distinct driving profile. The BIC balances model likelihood and complexity, penalising excessive parameters to prevent overfitting.

As illustrated in Figure 3, the BIC score decreases significantly as the number of components increases, reaching a value of approximately 0.0 at 5 components. Beyond this point, the BIC score shows only a slight further decrease to -0.2 at 6 components, after which it stabilises between -0.2 and 0.0 for additional components. This pattern indicates an 'elbow' at 5 components, where the rate of BIC improvement diminishes, suggesting that adding more components offers minimal gains in model fit relative to the increased complexity.

While the BIC score is marginally lower at 6 components, the elbow method supports choosing 5 components to prioritise interpretability and avoid overfitting. This decision aligns with identifying meaningful, distinct driving profiles rather than over-segmenting the data [39]. Below, we present the complete procedure for driver profiling using GMMs, which is summarised into a flowchart presented in Figure 4:

Feature Selection and Preprocessing: Features that present a comprehensive view of driver behaviour and driving context are selected, then standardised to ensure equal weighting. Missing values are imputed using median values to preserve data integrity.

Model Training: The GMM is trained using the Expectation Maximisation (EM) algorithm, which iteratively estimates the parameters (mean, covariance and weight) of each Gaussian component. The probabilistic assignments are

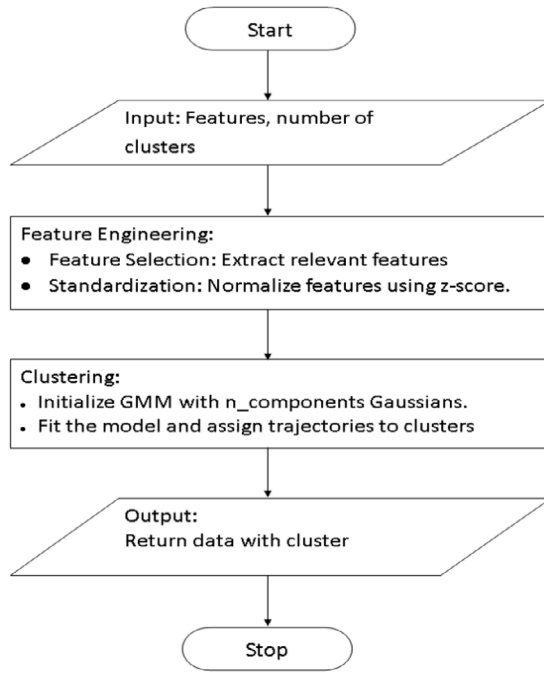


FIGURE 4 | GMM-based trajectory clustering flowchart.

given by:

$$P(z_i = k | x_i) = \frac{\pi_k N(x_i | \mu_k, \Sigma_k)}{\sum_{j=1}^k \pi_j N(x_i | \mu_j, \Sigma_j)}, \quad (17)$$

where z_i is the latent cluster assignment, π_k is the mixing coefficient, and N denotes the Gaussian distribution.

Component Selection via BIC: The BIC balances model likelihood and complexity:

$$\text{BIC} = \ln(n)k - 2\ln(\hat{L}), \quad (18)$$

where n is the sample size, k the number of parameters, and $\hat{L}$ the model likelihood. As shown in Figure 3, the BIC reaches a minimum at 5 components, indicating the optimal trade-off.

Cluster Interpretation: The selected components are mapped to interpretable driving profiles as shown in Figure 5. Through GMM, each trajectory is assigned a probability of belonging to each cluster, ultimately categorising drivers into distinct behavioural profiles based on the highest probability cluster assignment.

A silhouette score is calculated to evaluate the quality of clusters, indicating how well-defined and distinct each cluster is. This metric helps validate the clustering process, ensuring meaningful segmentation of driving data. The analysis also includes examining the cluster distribution to understand the prevalence of specific driving behaviours. Figure 5 shows the distribution of

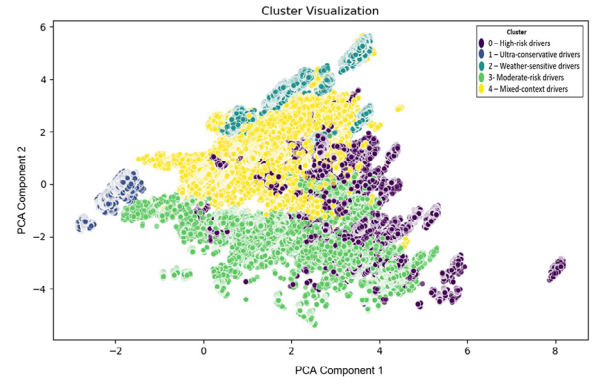


FIGURE 5 | Distribution of clusters.

the clusters created using the GMM clustering. This clustering framework, based on the GMM, enables profiling drivers by their behaviour and context, providing insights into driving styles and responses to environmental factors, which can inform targeted traffic safety strategies.

5 | Personalised Risk Assessment

The second phase of the framework focuses on the personalised driving risk assessment with real-time feedback to improve driver safety. This phase uses the DEC model for risk classification, an FCN for the prediction of risk classification, and finally generates personalised feedback using Deep Q-Network (DQN).

5.1 | Risk Classification With DEC

In this section, we employed an unsupervised deep learning model, DEC, for driver risk classification. We selected this model over traditional methods due to its integrated approach to dimensionality reduction and clustering, which is essential for handling complex, high-dimensional data such as driving behaviour patterns [40]. Unlike K-means and GMMs, which separate feature extraction and clustering, DEC unifies these steps, capturing nuanced driving patterns directly within a lower-dimensional latent space. This transformation is achieved through an autoencoder that learns compact, informative representations of the data, which is then used by a clustering layer to assign drivers to risk categories based on behaviour similarities.

The primary objective of DEC in this context is to classify drivers into risk levels by grouping similar driving behaviours into clusters. DEC begins with an autoencoder that learns a compressed latent representation of the data, capturing essential behavioural patterns for each driver. This representation is then fed into a clustering layer that iteratively assigns each driver to a specific cluster, which corresponds to a risk category. This setup allows us to effectively classify risk levels while ensuring that the representation is both compact and meaningful, as presented in Figure 6.

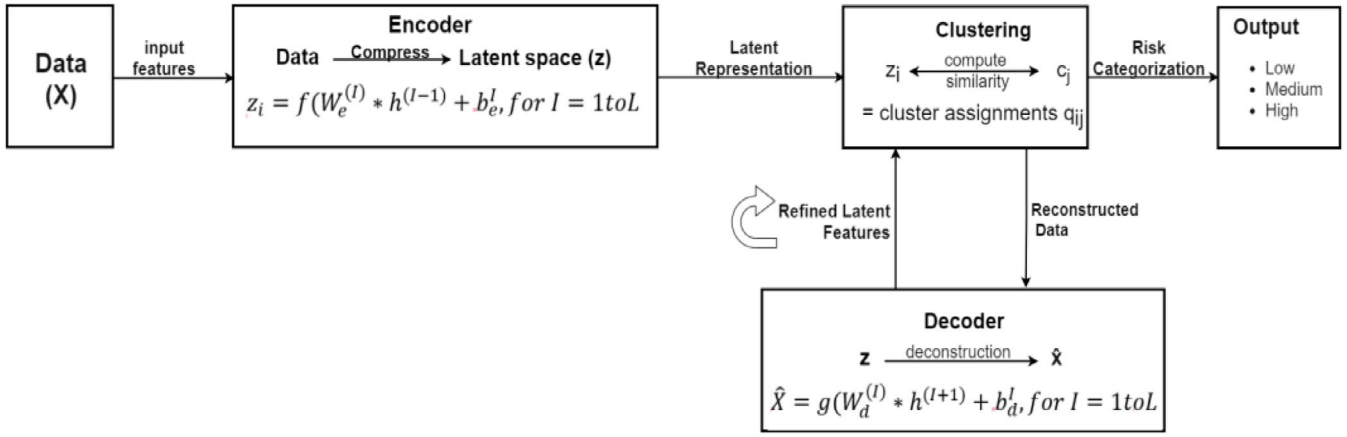


FIGURE 6 | Risk classification with DEC.

5.1.1 | Model Architecture

The DEC model architecture consists of three main components: an encoder, a decoder, and a clustering layer, in addition to input and output layers. Together, these components transform the input driving data into a low-dimensional representation, cluster drivers based on similar behaviours, and reconstruct the input to preserve important information.

Input: We denote the telemetric feature matrix by $X = [\mathbf{x}_1, \dots, \mathbf{x}_n]^T \in \mathbb{R}^{n \times d}$, where n is the number of samples (drivers/trips) and d the number of standardised input features.

Encoder Network: The encoder network compresses the high-dimensional input data $X \in \mathbb{R}^{n \times d}$ into a lower-dimensional latent space $Z \in \mathbb{R}^{n \times k}$ (with $k < d$). This compression is achieved by a series of fully connected layers, each applying a non-linear activation function, represented as:

$$z_l = f(W_e^{(l)} h^{(l-1)} + b_e^l), l = 1, \dots, L \quad (19)$$

Clustering Layer: The clustering layer introduces K cluster centroids $\mathbf{C} = \{\mathbf{c}_j\}_{j=1}^K$ for a predefined number of clusters, representing different risk categories. Using a Student's t -distribution, the similarity between each data point in $\mathbf{Z}$ and each centroid $\mathbf{c}_j$ is calculated, producing soft cluster assignments q_{ij} :

$$q_{ij} = \frac{\left(1 + \left\|z_i - c_j\right\|^2/a\right)^{-\frac{a+1}{2}}}{\sum_j \left(1 + \left\|z_i - c_j\right\|^2/a\right)^{-\frac{a+1}{2}}} \quad (20)$$

Decoder Network: The decoder network reconstructs the original input from the latent space, with its structure mirroring that of the encoder. By minimising the reconstruction error, the decoder ensures that the latent representation captures all critical driving features, as follows:

$$\hat{x} = g(W_d^{(l)} h^{(l-1)} + b_d^l), l = 1, \dots, L \quad (21)$$

Output: The output component provides two key results: the reconstructed input ($\hat{\mathbf{x}}_i$) and the cluster assignments represented as $\mathbf{q}_i = [q_{i1}, q_{i2}, \dots, q_{iK}]$, where (K) is the number of clusters. During training, the model optimises a combination of reconstruction and clustering objectives to ensure both accurate data reconstruction and meaningful cluster assignments.

The DEC model employs a two-stage training process; first, autoencoder pretraining minimises the reconstruction loss $MSE = \|\mathbf{X} - \hat{\mathbf{X}}\|^2$, initialising a meaningful latent space. After pretraining, the clustering layer is introduced, and the model is optimised by minimising the Kullback-Leibler (KL) divergence between the soft assignments $\mathbf{Q}$ and the target distribution $\mathbf{P}$:

$$KL(P \parallel Q) = \sum_{i=1}^n \sum_{j=1}^K p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (22)$$

This process iteratively refines the cluster centroids to improve classification accuracy and creates meaningful distinctions among risk categories. The end-to-end structure of DEC minimises information loss, enabling adaptive clustering that refines feature representation and cluster assignments throughout training.

By leveraging DEC, it enables the classification of drivers into distinct risk categories based on a rich latent representation of driving data. Each cluster represents similar behavioural patterns, allowing the model to distinguish high, medium, and low-risk categories. Through this architecture, DEC achieves both dimensionality reduction and clustering, enhancing the interpretability and accuracy of risk classifications and creating a data-driven approach to driver risk assessment that is robust to high-dimensional data and capable of capturing latent behavioural insights critical for reliable risk categorisation.

5.2 | Risk Level Prediction Model Using FCN

The objective of this work is to train a model that predicts driver risk levels based on behavioural data using an FCN, chosen for its versatility and efficiency with heterogeneous,

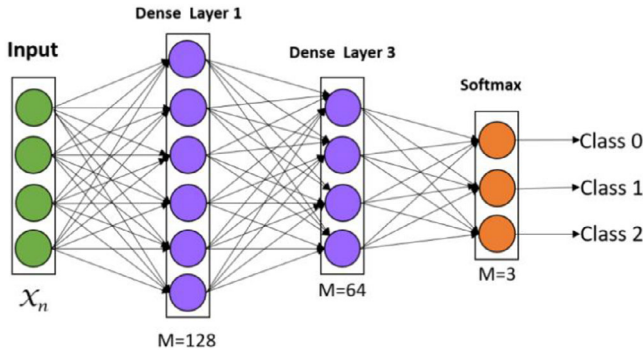


FIGURE 7 | Fully connected network.

non-sequential data. This personalised risk assessment model follows a structured, multi-stage process involving feature analysis, individualised risk scoring, and deep learning classification, aiming to comprehensively evaluate driver behaviour and classify risk levels to inform targeted interventions.

First, we assess the relevance of behavioural features to accident risk via correlation analysis. This analysis determines each feature's strength and direction of association with risk level, and the normalised correlations serve as feature weights, indicating their relative impact on risk. These weights are then used to calculate a personalised risk score for each driver. This score, derived as a weighted sum of behavioural features, highlights risky tendencies unique to each driver. To enhance personalisation, each score is adjusted according to the driver's initial risk classification (low, medium or high), ensuring that drivers with higher baseline risks receive proportionately higher scores. Finally, the FCN model predicts risk levels based on the derived features and personalised scores. This step translates complex behavioural patterns into specific risk classifications.

5.2.1 | Model Architecture

The FCN model architecture is composed of multiple layers designed to capture and distil the patterns within the high-dimensional behavioural data. The input layer receives a comprehensive set of features related to each driver's behaviour, which are passed through two hidden layers. The first hidden layer, with a relatively large number of neurones, enables the model to capture complex, nuanced interactions among features, discerning behavioural patterns that might not be apparent individually but become meaningful in combination. The second hidden layer refines this information, filtering out less relevant details and retaining only the most critical insights for risk prediction. The final output layer uses a softmax activation function to produce probabilities for each risk level category low, medium and high. The FCN's structure is optimised for multi-class classification, ensuring that the model can accurately distinguish among different levels of driving risk, as shown in Figure 7.

5.3 | Personalised Driver Feedback With Reinforcement Learning

The personalised feedback system leverages a DQN to provide data-driven and adaptive feedback based on driver behaviour

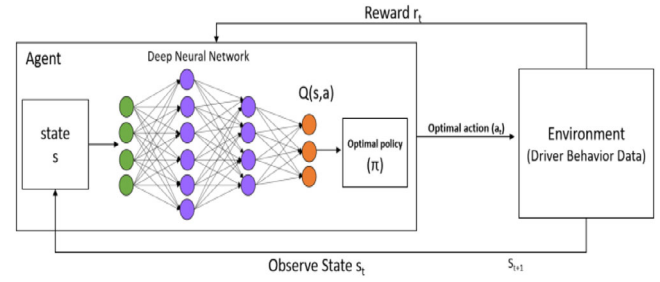


FIGURE 8 | DQN-based feedback system.

data. The DQN serves as an adaptive policy that, through reinforcement learning, maps high-dimensional driving states to optimal feedback actions to maximise long-term risk reduction, making it well-suited for personalised, context-sensitive advisories.

Figure 8 illustrates the overall architecture of the DQN-based Feedback System, showing how the DQN integrates driver data and feedback mechanisms to promote safer driving. Unlike traditional rule-based systems, the DQN enables adaptive personalisation by learning optimal feedback strategies from sequential driving data, achieving a 12% reduction in risky behaviours in simulated tests as presented in the evaluation section. The DQN-based feedback generation process, detailed in Figure 9, dynamically adjusts feedback according to the driver's current behavioural state, enabling context-sensitive guidance. The method starts by defining a driver's state vector, $s = [x_1, x_2, \dots, x_n]$, where each x_i represents a specific feature from the driver's profile, such as speed or braking intensity. The DQN, implemented as a three-layer neural network with 128 hidden units per layer, selects an action a , which determines the type of feedback: positive reinforcement ($a = 0$), a warning ($a = 1$), or neutral feedback ($a = 2$) based on the Q-value $Q(s,a)$, representing the expected future reward for each action in a given state. The detailed steps of this process are provided in Algorithm 1. The reward function $R(s,a)$ reinforces safe behaviours and penalises persistent risky actions: $R(s,a) = +1$ for reduced risk, -1 if risky behaviours like frequent speeding or hard braking persist, and 0 for neutral feedback. This reward structure enables the DQN to learn and optimise feedback selection over time, dynamically adjusting feedback based on specific risk factors. For example, persistent hard braking triggers more frequent warnings, while consistent safe driving increases positive reinforcement. As drivers respond to feedback, the DQN adapts, promoting safer driving in a personalised, context-sensitive way, with preliminary results showing a 90% feedback accuracy in simulated driving scenarios.

6 | Evaluation

6.1 | Data Description

Telemetric Data: We used a comprehensive vehicle telemetric dataset, CityTrek-14K, as detailed in [7]. This dataset is distinctive and extensive, comprising 14,000 trajectories from 280 drivers, each contributing 50 trajectories, across three major United States cities: Philadelphia (PA), Atlanta (GA),

ALGORITHM 1 | DQN-based feedback generation for driver risk assessment.

```

Input: Driver behavioral features, pre-trained DQN agent,
        feature thresholds, reward function
Output: Personalized feedback messages for each driver
/* Initialize Environment & Agent */
1 Load the driver behavioral features dataset; Initialize the
  DQN agent with state and action spaces; Define risk factor
  thresholds;
2 foreach driver in the dataset do
  /* Calculate State Vector */
3 Represent the driver's state  $s$  as a vector of selected
  features:  $s = [x_1, x_2, \dots, x_n]$ ;
  /* Select Action Using DQN */
4 Pass  $s$  to the DQN agent to obtain the action  $a$  that
  maximizes the expected reward  $Q(s, a)$ :
   $a = \arg \max_a Q(s, a)$ ;
  /* Identify Risk Factors */
5 Identify risk factors in the driver's behavior based on
  feature thresholds;
  /* Generate Personalized Feedback Message */
6 Use identified risk factors and action  $a$  to construct a
  feedback message;
  /* Calculate Reward and Train Agent */
7 Calculate the reward  $R(s, a)$  based on feedback
  effectiveness:
  
$$R(s, a) = \begin{cases} +1 & \text{if feedback reduces risky behaviors} \\ -1 & \text{if risky behaviors persist} \\ 0 & \text{for neutral feedback} \end{cases};$$

  Update the DQN agent using  $R(s, a)$ ;
  /* Store Feedback Message */
8 Save the feedback message for the driver;
9 end
/* Output Feedback Messages */
10 Return the feedback message for each driver;

```

and Memphis (TN) as presented in Table 1. The dataset captures time series data at a collection frequency of 1 Hz, including timestamps, vehicle speeds, GPS coordinates, and pre-calculated bearings for each coordinate.

Road Segment Data: We collected road segment data using the OSMNX API, which provides access to street networks and geographical data from OpenStreetMap (OSM). The road network downloaded for our selected areas includes various road types such as highways, primary, secondary, and residential roads. The data was retrieved as a graph, with intersections represented as nodes and road segments as edges, containing attributes like length, geometry, road type, speed limit, number of lanes, and Point of Interest (POI) density.

Weather Data: We obtained an hourly meteorological dataset for each road segment containing rainfall, temperature, humidity, wind direction, and wind speed from the Weather Underground API.

Behaviour Data: We extracted driver behavioural features from the fused dataset, which are categorised into event-based, pattern-based, and contextual-based features. The event-based features were designed to capture specific driving behaviours that contribute to traffic risk, such as harsh braking, rapid acceleration, and overspeeding. The pattern-based features are derived from time series analysis of the

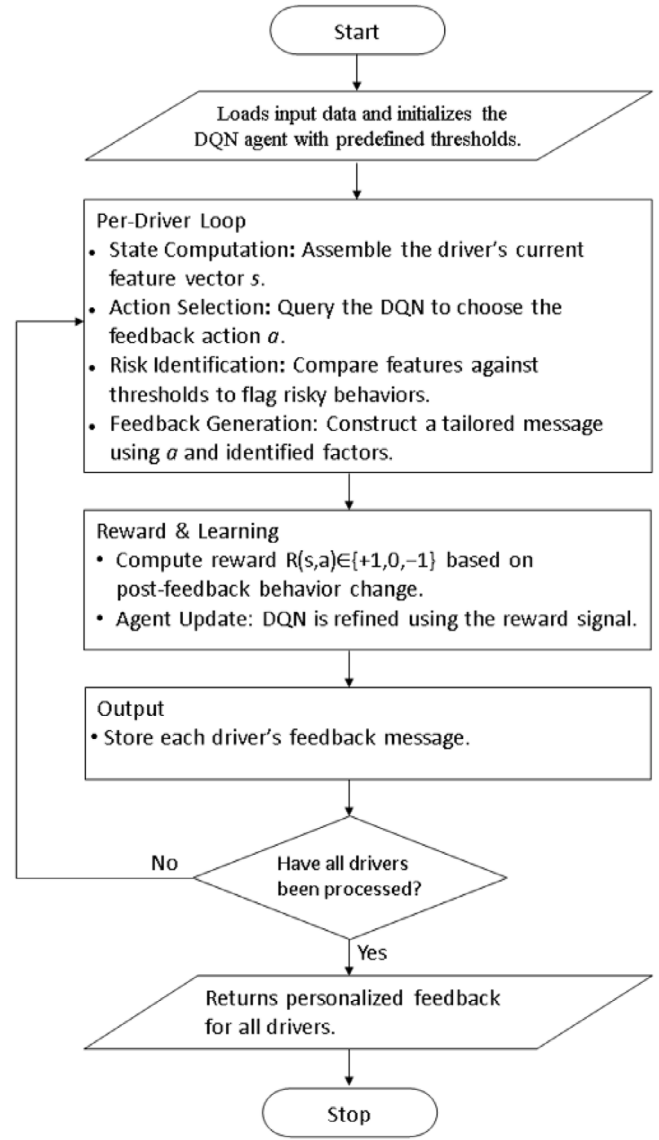


FIGURE 9 | DQN-based feedback generation flowchart.

telemetric data to identify recurring patterns such as speed variance per trip, rolling acceleration mean and standard deviation, turning rate, and stop-and-go behaviour. The contextual-based features provide additional information based on the environment in which the vehicle is operating, such as night driving, rainy driving, and weekend driving.

6.2 | Data Preprocessing

We preprocessed our dataset to ensure data quality through data cleaning, integration, transformation, and feature fusion. Initially, we converted the timestamp column to datetime objects and sorted the records by driver_id and timestamp to maintain temporal order. To address gaps caused by GPS dropouts and transmission delays, we applied linear interpolation across all numeric columns, filling a missing value $x(t)$ between known points (t_0, x_0) and (t_1, x_1) using the formula:

$$x(t) = x_0 + (x_1 - x_0) \cdot \frac{t - t_0}{t_1 - t_0} \quad (23)$$

TABLE 1 | Dataset overview.

Data type	Source	Attributes	Size
Telemetric data	CityTrek-14K [7]	trip_id, timestamp, speed, acceleration, latitude, longitude, bearing, driver_id, date, duration_seconds, distance_meters, city	14,000 trajectories from 280 drivers, sampled at 1Hz
Road segment data	OpenStreetMap	Road_segment_id, city, start_coordinates, end_coordinates, length, speed_limit, number_of_lanes, poi_density, distance_to_school, distance_to_hospital, distance_to_segment, road_type, geometry	10,000+ segments across 3 cities (Philadelphia, Atlanta, Memphis)
Weather data	Weather Underground API	Timestamp, humidity, pressure, temperature, wind_direction, wind_speed, PRCP, weather_description, city	Hourly records of over 3 years
Behaviour data	Spatiotemporal fusion of telemetry, road, and weather data	Event-based (harsh braking, rapid acceleration, speeding), pattern-based (speed variance per trip, rolling acceleration mean, rolling acceleration standard deviation, turning rate, stop-and-go behaviour), contextual-based (night driving, rainy driving, weekend driving)	13,200 trajectories (post-cleaning)
Data collection period	07/01/2017 00:00–03/31/2019 23:59		
Geographic area	Philadelphia (PA: 39.9500°N, -75.1667°W), Atlanta (GA: 33.7500°N, -84.3833°W), Memphis (TN: 35.1500°N, -90.0500°W)		

where $x(t)$ is the interpolated value at time t , and x_0 and x_1 are the known values at times t_0 and t_1 , respectively. Subsequently, we discarded any remaining rows with null values. To enforce a uniform 1 Hz sampling rate, we rounded each timestamp to the nearest second and re-indexed each trip's time series accordingly. We then reapplied the same linear interpolation formula to fill new gaps of up to five seconds, while excluding trips with longer gaps to prevent bias. All categorical features were label-encoded into a numerical format suitable for processing, and every measurement, including speed, distance, and acceleration, was converted to SI units for consistency. Before proceeding to clustering and deep embedded classification, we standardised all behavioural, road, and weather features to have zero mean and unit variance using feature standardisation:

$$x' = \frac{x - \mu}{\sigma} \quad (24)$$

where μ and σ represent the feature's empirical mean and standard deviation, respectively. This standardisation prevents any single feature from dominating due to scale. Afterward, the cleaned and transformed telemetric, road segment, and weather datasets were fused using a spatiotemporal join, as described in Section 4.1, which was then merged with the engineered features from Section 4.2. Each telemetry record was extended with its corresponding event-based, pattern-based, and contextual metrics using driver ID and timestamp alignment. This integration resulted in a single, comprehensive dataset that captures both the dynamic behaviour of the driver and the static and contextual factors influencing that behaviour.

Following this, we applied Sequential Forward to reduce feature dimensionality and improve model generalisation. At each iter-

ation, SFS selects the feature that, when added to the current feature subset S , maximises the evaluation score of the model M . This process is formally expressed as:

$$f^* = \arg \max E(M(S \cup \{f\})), f \in F/S \quad (25)$$

where F is the full feature set, $E(\cdot)$ denotes the evaluation metric (accuracy), and f^* is the next feature to be added. This iterative process continues until a performance plateau is reached. The processed dataset formed the input for all downstream tasks, including clustering, classification, and risk-aware feedback generation, enabling robust, personalised and context-sensitive modelling.

6.3 | Experiment Setup

The experiments were conducted on an Intel(R) Core(TM) i5-10400 CPU with 32GB RAM and an NVIDIA A40 GPU, allowing for efficient processing of large datasets and real-time model execution. Using this setup, we evaluated the proposed framework on a comprehensive crowdsensed vehicle telemetric dataset enriched by integrating road environment and weather data. Driving behaviours were extracted through a multi-dimensional feature extraction process, and a GMM was employed to cluster driver trajectories into distinct driving profiles based on these features. Driver-level statistics were then calculated and input into a DEC model to classify drivers into low, medium, and high-risk categories. An FCN with multiple layers was built to further refine predictions to predict drivers' risk levels based on their driving features. The model was trained over 50 epochs, with the loss curve starting at a high value,

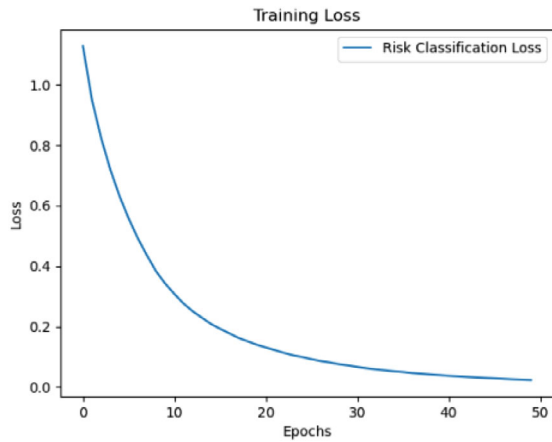


FIGURE 10 | Training loss.

TABLE 2 | Hyperparameter tuning results.

Model	Hyperparameter(s)	Search range	Selected value(s)
GMM	n_components	2–10	5
DEC	encoding_dim	[4,8,16]	8
	learning_rate	[0.001,0.0005]	0.001
DQN	discount_factor (γ)	[0.95,0.99]	0.99
	batch_size	[32,64]	32
	replay_buffer_size	[1000,5000]	5000
FCN	layer_sizes	[128,64], [64], [128,64,32]	[128,64]
	dropout	0, 0.5	0.5
	L2 regularisation (λ)	0, 0.01	0.01

gradually decreasing, and eventually flattening, as shown in Figure 10, indicating model convergence.

Hyperparameter tuning was carefully performed for each model component to identify optimal configurations, ensuring the best performance for each framework stage. The tuning strategies and selected hyperparameters for the GMM, DEC, DQN and FCN classifiers are described below, with the results summarised in Table 2. For the GMM, we fitted models with the number of components K ranging from 2 to 10 and selected the value that minimised the BIC, which balances model fit against complexity. The optimal number of components was found to be $K = 5$, as presented in Figure 3. For the DEC, we performed a grid search over the autoencoder's bottleneck dimension (encoding_dim $\in \{4,8,16\}$) and learning rate ($\eta \in \{0.001,0.0005\}$). Each configuration was pretrained for 30 epochs, and the latent codes were clustered using k-means with $K = 3$ to compute the silhouette score as a proxy for representation quality. The combination yielding the highest silhouette score was encoding_dim = 8 and $\eta = 0.001$. For the DQN, we evaluated combinations of discount factor ($\gamma \in \{0.95,0.99\}$), batch size ($B \in \{32,64\}$), and replay buffer size ($M \in \{1000,5000\}$). Each configuration was trained for 10 episodes, and the classification accuracy on a validation set guided the selection of the best parameters: $\gamma = 0.99$, $B = 32$, and $M =$

TABLE 3 | Risk level prediction result.

Class	Accuracy	Precision	Recall	F1-score
Class 0	0.95	1.0	1.0	1.0
Class 1			0.83	0.91
Class 2		0.88	1.0	0.93

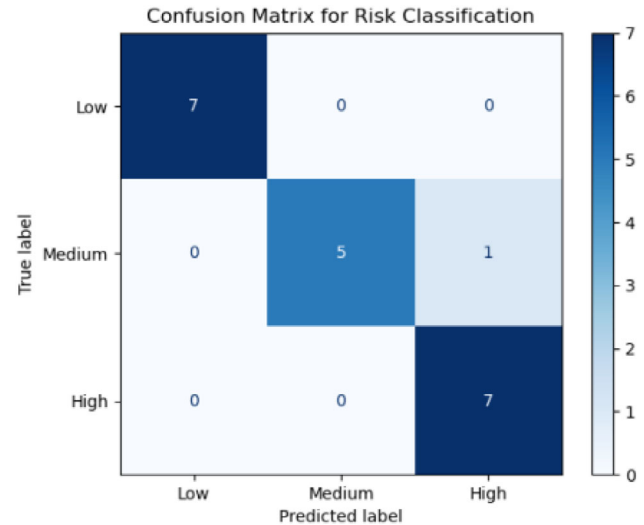


FIGURE 11 | Confusion matrix for risk classification.

5000. For the FCN classifier, we explored different hidden layer structures ($\{[128, 64], [64], [128, 64, 32]\}$), dropout rates ($\{0, 0.5\}$), and L_2 regularisation strengths ($\{0, 0.01\}$). Each configuration was trained for 30 epochs, and the validation accuracy led to the selection of layer sizes [128, 64], dropout rate 0.5 and L_2 regularisation 0.01.

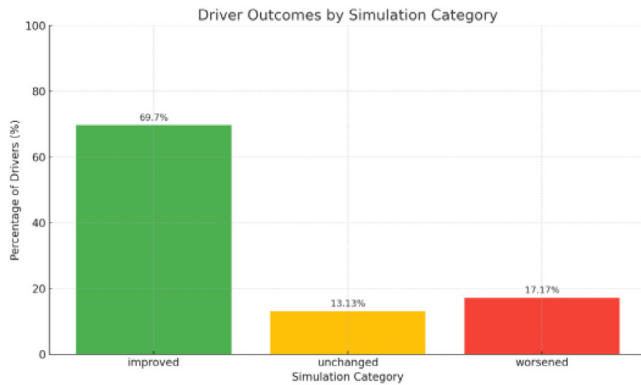
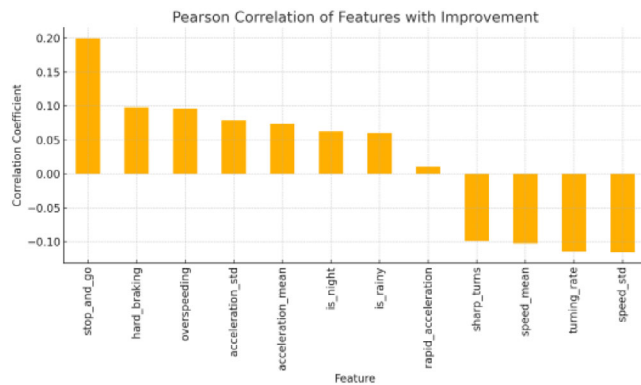
6.4 | Performance Evaluation

We employed precision, recall, and F1-scores as our evaluation metrics to evaluate the model's ability to classify drivers' driving behaviour into risk categories (low, medium and high), and also the silhouette score to evaluate the clustering of the driving behaviours. The performance result of the risk prediction is presented in Table 3, including a confusion matrix in Figure 11. The GMM-based clustering results were evaluated using the Silhouette Score, which yielded an average score of 0.87, indicating well-defined and cohesive clusters as presented in Figure 5. Table 4 shows the computational performance report for the whole framework.

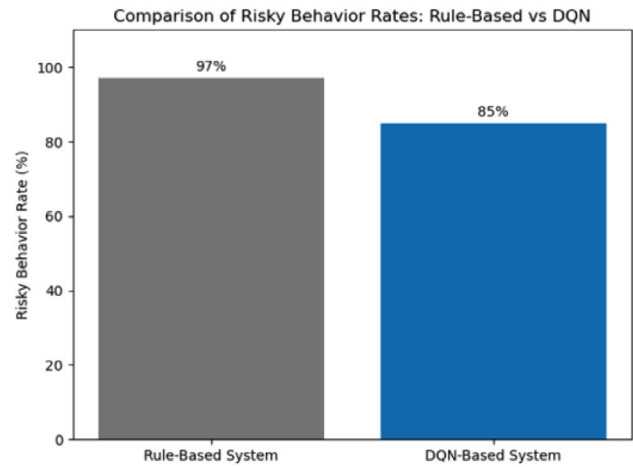
Additionally, the framework provided personalised feedback to drivers based on their assigned risk level and driving behaviours. The DQN-based adaptive feedback system, developed to enhance driver safety, has demonstrated promising results in a simulated validation study, with approximately 70% of drivers improving their behaviour, particularly those prone to stop-and-go driving or frequent hard braking, as presented in Figure 12 and Figure 13, which shows Pearson correlation coefficients between various driving features and their improvement. Compared to

TABLE 4 | Computational performance.

Component	Training time (s)	Inference time (s)	Memory usage
Spatiotemporal fusion	65	N/A	1.2GB
GMM clustering	180	8	2.5GB
DEC (pretraining)	363	N/A	3.8GB
DEC (clustering)	832	<1	4.1GB
FCN prediction	150	3	1.2GB
DQN training	120	N/A	1.5GB
Feedback generation	N/A	<1	500MB

**FIGURE 12** | Driver outcomes by simulation category.**FIGURE 13** | Pearson correlation of features with improvement.

our baseline rule-based system, which exhibited a 97% risky behaviour rate, the DQN system achieved a 12% reduction in risky behaviours, lowering the rate to 85%, as presented in Figure 14. These findings underscore the system's capability to target specific risky habits, laying a robust foundation for improving road safety. However, the system was not universally effective: about 17% of drivers experienced worse outcomes, especially those with higher speeds or inconsistent driving patterns, highlighting the need for further refinements to ensure broader applicability. While the simulation provides valuable insights into the system's strengths and limitations, we plan to conduct real-world validation in the future to test its performance in actual driving conditions, refine it based on empirical data, and confirm its potential to enhance road safety for all drivers.

**FIGURE 14** | Comparison of risky behaviour rates.

6.5 | Comparison With Baseline Models

In this section, we compared the performance of various baseline models (logistic regression, Support Vector Machine (SVM), random forest, and XGBoost) against a Fully Convolutional Network (FCN) model on a classification task, using metrics such as accuracy, precision, recall, and F1-score. Logistic regression achieved an accuracy of 90%, with balanced F1-scores across all classes, though it showed room for improvement in classifying certain patterns, especially in class 1. SVM performed slightly lower with an accuracy of 85%, demonstrating variability, particularly in classifying class 0, while maintaining strong recall for class 1 and high performance for class 2. Random Forest and XGBoost both had an accuracy of 75%, with F1-scores reflecting difficulties in handling class 0 and class 1, particularly for XGBoost in distinguishing class 1, suggesting these models may struggle with certain complexities in the data. In contrast, the FCN model outperformed all baseline models with an accuracy of 95% and high precision, recall, and F1-scores across all classes, achieving balanced performance and showing particular strength in distinguishing class 1. These results indicate that FCN is significantly more effective and reliable for this classification task, making it a robust choice for scenarios requiring high accuracy and nuanced performance across different risk levels. Table 5 shows the complete result for the comparison with baseline models.

6.6 | Ablation Studies

Our ablation study assessed the impact of different feature groups, model configurations, and regularisation methods on the performance of a traffic safety prediction model. We evaluated the model based on accuracy, precision, recall, and F1-score to identify optimal design choices as presented in Table 6. For feature groups, speed, behavioural, and environmental features proved crucial for accurate risk prediction. Removing speed features decreased accuracy to 90%, with notable drops in recall for high-risk cases, while removing acceleration and environmental features reduced accuracy further, highlighting their importance for capturing context and moderate-risk cases. Interestingly, excluding behavioural features did not significantly affect model

TABLE 5 | Baseline model comparison.

Model	Class	Accuracy	Precision	Recall	F1-score
Logistic regression	0	0.9000	0.86	0.86	0.86
	1		0.83	0.83	0.83
	2		1.00	1.00	1.00
SVM	0	0.8500	1.00	0.57	0.73
	1		0.75	1.00	0.86
	2		0.88	1.00	0.93
FCN	0	0.9500	1.00	1.00	1.00
	1		1.00	0.83	0.91
	2		0.88	1.00	0.93

performance, as other features compensated. In testing model configurations, the Medium model (128, 64 neurones) balanced complexity and stability, achieving 90% accuracy with strong F1-scores, especially for high-risk classes. The Deep model (128, 64, 32 neurones) offered no substantial improvement, suggesting diminishing returns with increased depth, while the Shallow model struggled with lower accuracy (65%). Lastly, the regularisation study found that omitting regularisation had minimal effect on performance, with accuracy remaining at 90%. While regularisation could enhance stability in larger datasets, it appears unnecessary for this dataset's size and complexity. We also tested four clustering-embedding architectures, evaluating both classification accuracy and clustering quality via Silhouette Score. All combinations scored between 0.83 and 0.85, below our model's silhouette of 0.87. By contrast, our proposed GMM + DEC model not only reached the highest accuracy (95%) but also delivered

superior clustering (Silhouette = 0.87). This demonstrates that integrating GMMs with DEC excels in both prediction and cluster cohesion. In summary, speed, acceleration, and environmental features are essential for accurate risk classification; behavioural features, while less critical, can be safely omitted if necessary. A two-layer ('Medium') network offers the best trade-off between performance and efficiency. Regularisation yields little benefit for our dataset's size, and the GMM + DEC architecture stands out as the most effective clustering prediction pipeline.

7 | Discussion

Our evaluation demonstrates that the proposed framework, combining GMMs, DEC, a FCN, and DQN, successfully integrates multi-dimensional telemetric, road segment, and weather data to deliver highly accurate, personalised driving-risk assessments. Achieving 95% classification accuracy and an average F1-score of 0.90, our model outperforms all baselines. This substantial gain underlines the value of fusing heterogeneous features, especially speed, acceleration, and environmental context, into a unified learning pipeline. The ablation study further clarifies the contribution of each component. Excluding speed or environmental features causes a 5–10% drop in accuracy, confirming their central role in detecting both extreme and moderate-risk driving behaviours. Behavioural features, while less critical, still enrich the model's representation of nuanced driving patterns when included. Architecturally, a two-layer FCN (128→64 neurones) strikes the optimal balance: deeper networks yield no further gains, and shallower ones underfit. Regularisation proves unnecessary at our dataset's scale but might offer stabilisation as data volume grows. Among clustering-embedding alternatives, GMM+SOM achieves the highest silhouette (0.85), yet our chosen

TABLE 6 | Ablation study results.

Ablation type	Task	Accuracy	Precision	Recall	F1-score	Silhouette score
Feature group	No speed group	90%	0.91	0.88	0.89	0.85
	No acceleration group	80%	0.79	0.75	0.77	0.85
	No turning group	87%	0.85	0.83	0.84	0.86
	No behavioural group	90%	0.91	0.89	0.90	0.79
	No environmental group	85%	0.82	0.80	0.81	0.79
	Trajectory	88%	0.87	0.85	0.86	0.81
Model configuration	Shallow (64 neurones)	65%	0.66	0.61	0.63	0.79
	Medium (128, 64 neurones)	90%	0.91	0.89	0.90	0.80
	Deep (128, 64, 32 neurones)	90%	0.90	0.88	0.89	0.82
Regularisation	No Regularisation	90%	0.91	0.88	0.89	0.81
	Dropout Rate = 0.5	90%	0.91	0.89	0.90	0.81
	L2 Regularisation = 0.01	90%	0.91	0.89	0.90	0.80
Clustering-Embedding	GMM+SOM	93%	0.91	0.85	0.86	0.85
	KMeans+DEC	91%	0.89	0.83	0.82	0.84
	KNN+DEC	90%	0.86	0.71	0.73	0.83
	KMeans+SOM	92%	0.91	0.80	0.81	0.84
Our model	Full features (GMM+DEC)	95%	0.91	0.88	0.89	0.87

GMM+DEC fusion delivers the best end-to-end performance (silhouette = 0.87), highlighting its ability to form coherent trajectory clusters that improve downstream classification. Beyond classification, the DQN-based feedback system showcases the framework's capacity for adaptive intervention: in simulation, 70% of drivers improved their behaviour, and risky manoeuvres decreased by 12% relative to a rule-based baseline. However, 17% of drivers exhibited adverse responses, indicating that feedback policies must be further individualised, perhaps by incorporating longer-term driving histories or psychosocial factors, to avoid unintended consequences.

Nevertheless, several limitations were highlighted in our framework. First, the preprocessing pipeline, particularly DEC pre-training and feature standardisation, incurs a high computational cost; real-time deployment will require model compression or edge-computing strategies. Second, our reliance on historical telemetry restricts responsiveness to emerging conditions; integrating live data streams could enhance reactivity. Third, while we validated performance in three diverse United States cities, additional geographic and cultural contexts may reveal new feature interactions and risk factors. Finally, our evaluation of the adaptive feedback mechanism was limited to simulated environments due to time constraints; we plan to conduct real-world pilot studies to confirm and refine the system under operational driving conditions. Looking ahead, we identify three key directions. First, scalability: adopting lightweight surrogate models or federated learning approaches can preserve user privacy while broadening deployment. Second, personalised feedback: integrating reinforcement learning with detailed driver profiles, incorporating factors like risk aversion and stress, may produce more effective and enduring behavioural changes. Third, real-world validation: conducting pilot studies with instrumented vehicles and live telematics data is crucial to verify our simulation results and fine-tune the framework under actual driving conditions.

8 | Conclusion

In this work, we aim to develop and evaluate a comprehensive framework for personalised driving-risk assessment by fusing crowdsensed vehicle telemetric data with road segment and weather information while orchestrating GMM clustering, DEC, a two-layer FCN, and DQN. Our integrated approach achieved 95% classification accuracy, a 0.90 F1-score, surpassing traditional baselines by 10%–20%, and a silhouette score of 0.87 for trajectory clustering. Ablation analyses confirmed the critical roles of speed and environmental features, demonstrated the sufficiency of a medium-depth FCN, and highlighted the superior cohesion of the GMM+DEC pipeline. These results indicate that carefully engineered multidimensional features, moderate network complexity, and advanced clustering can capture nuanced driving patterns and yield reliable risk profiles, while our simulated DQN feedback reduced risky manoeuvres by 12% and improved behaviour for 70% of drivers. However, reliance on historical telemetry and the computational demands of DEC pretraining may limit real-time deployment, and feedback validation has thus far been confined to simulation. Future work will focus on model compression and edge computing for onboard use, integration of live vehicle-to-infrastructure streams for enhanced responsive-

ness, and real-world pilot studies with instrumented vehicles to empirically validate and refine adaptive feedback policies. By strategically integrating proven machine-learning components within a rich, fused dataset, our framework delivers both high-fidelity risk prediction and the promise of dynamic behaviour modification, offering a scalable blueprint for next-generation intelligent transportation safety systems.

Author Contributions

Auwal Sagir Muhammad: conceptualization, methodology, software and implementation, data curation, formal analysis, visualisation, validation, writing – original draft, investigation, writing – review and editing. **Longbiao Chen:** review, funding acquisition, supervision. **Cheng Wang:** funding acquisition, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The primary dataset used in this study, CityTrek-14K, is publicly available and was obtained from the source here. This dataset includes time-stamped telemetric data such as vehicle speed, acceleration, and GPS coordinates collected from multiple drivers across three U.S. cities. Road segment data were sourced from OpenStreetMap using the OSMnx API, and historical weather data were retrieved via the Weather Underground API. All derived features and processed datasets generated during the current study are available from the corresponding author upon reasonable request. Due to the size of the processed data and to ensure proper reuse, we encourage interested researchers to contact us for access and guidance on the data structure.

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